

2. FilesystemBackend

FilesystemBackend is the opposite of the StateBackend.

It need the root directory given by the developer so that it can operate on real files under the path.

FilesystemBackend reads and writes the files under a configurable root directory on the local machine.

By specifying the root, the agent has access to everything under it.

Best use case: Coding Assistant, local development CLIs

```
from deepagents import create_deep_agent
from deepagents.backends import FilesystemBackend

agent = create_deep_agent(
    backend=FilesystemBackend(root_dir=disk_path.resolve(), virtual_mode=True)
)
```

`virtual_mode = True` let the agent read/write real files on disk, but confine it strictly inside `disk_path` — no escaping via `..`, symlink tricks, or absolute paths." This is the standard, recommended pattern any time you give a deep agent real filesystem access (as opposed to `virtual_mode=False`, which is basically "give it your whole machine").

Comparison

Backend	Lives in	Cross-thread?	Survives restart?	Real file on disk?
StateBackend	LangGraph state	✗	✗	✗
FilesystemBackend	Your disk	✓	✓	✓
StoreBackend	A LangGraph store	✓	only w/ persistent store	✗

-